Week41- Assignment-Solution

Task1: Steering behaviors

Take three steering behaviors from this week's lecture and- explain how they work interms of the directions of the forces and what they achieve.

Seek & Flee:
Seek moves straight towards the target. Does not slow down as it approaches the target and seek does not take into account maximum power / speed.
Flee e r the opposite of seek and generates a force that moves the agent directly away from the target position as quickly as possible.

Pursue & Evade:
Pursue is useful for catching or hitting a moving target. It tries to calculate where the target will be and then aims .
Evades the opposite of Pursue : Generating r a force that moves the agent directly away from the target's predicted next position as quickly as possible.

Arrive:
Arrive t he height to slow down as it approaches the target. Checks the remaining distance to find the correct speed.

Task 2: Prototyping Online Components

a) What are the benefits of letting your own IoT dings or app connect to a dedicated service that takes care of communication with 3rd party solutions? (Unlike letting the dings / app talk to 3rd party solutions directly.)

If the third party solution changes its system, the IoT thing can stop working unless all the devices are updated. By going through our own service it means that we make adjustments in the service - the communication between IoT dings and service can remain unchanged.

b) API and scraping are two ways we can retrieve data from 3rd party system. Say a little about how each of these works and which variant is preferable (and why) if we can use both.

API: Application Programming Interface (API) is recommended to use. An API f or third-party system is a set finished methods / functions that can be called on to retrieve information from third-party system.

scraping: If there is no API, we can retrieve information from a third-party system by reading the content of the website / solution and extract the information we are interested in or take a screenshot and run character recognition to retrieve the information. This method is more difficult to implement, slower and more fragile than using an API. API is therefore preferred. NB: Remember to check licenses / rights against third parties to see if you are allowed to collect data from them in this way!

c) If we are going to build our own communication solution- e.g. ifbm. own service, say a little about what standards you want to choose and why.

One neat and easy way to communicate with a service is REST, via http. Here you get a lot for free in the form of existing protocols. There is no need at all to think about packages and data

transfer issues. Easy for multiple things / apps to connect to at the same time. REST primarily uses the GET, POST, PUT and DELETE methods. Each command that one executes becomes a unique URL. You can use JSON or XML to transfer the data.